

CSCE 650 VIRTUAL REALITY · MODULE 13 · LECTURE NOTES

Designing and Evaluating VR

Conventions, accessibility, and finding out whether it works.

MODULE OUTCOMES

- Apply the established VR interface conventions.
- Design for accessibility in VR specifically.
- Run an honest user evaluation.
- Choose measures appropriate to a perceptual claim.
- State the ethical considerations the medium raises.

1 Conventions

Convention	Why it settled	Cost of violating it
World-space UI at 1–2 m	A comfortable vergence distance near the display's focal plane (Module 02), and it can be occluded correctly (Module 03).	Eye strain, and an interface that fights the depth cues.
Grip to grab, trigger to select	Near-universal across applications and platforms.	Users will try the expected mapping first and conclude the application is broken.
Teleport on the thumbstick, as the default	Expected by anyone who has used VR before, and the safe choice for anyone who has not (Module 08).	A novice's first experience is a smooth-locomotion one, which is the worst possible introduction.
Menus attached to a hand or wrist	Always reachable, never in the way, and they move with the user.	A menu fixed in the world is lost the moment the user turns around.
Proximity highlighting	Teaches interactivity without words (Module 09).	Users do not discover what is interactive.
A visible guardian boundary	Safety — and users come to rely on it to know how far they can move.	People walk into walls. This is not hypothetical.

Conventions are worth following unless you have a specific reason not to. Novelty in interaction is paid for by the user, not by the designer — every unfamiliar mapping is a thing someone must discover, and the discovery happens in the first minute when they are least able to afford it.

COMFORT SETTINGS ARE AN ACCESSIBILITY FEATURE

Given the variation documented in Module 07 — where some people tolerate smooth locomotion indefinitely and others cannot manage two minutes — **an application with a single locomotion scheme has excluded part of its audience by construction.**

The standard set: teleport and smooth locomotion, both offered; snap and smooth turning, with snap as the default; adjustable vignette strength; seated and standing modes; and dominant-hand selection.

And height calibration. Assuming a standing adult of average height excludes children, wheelchair users, anyone sitting on a sofa, and anyone unusually short or tall. An adjustable height offset is a few lines of code.

These are cheap to implement at design time and expensive to retrofit, because locomotion choice propagates into level design (Module 08). **Which is why they belong in the design from the start** rather than in a patch after the complaints arrive.

- **Seated play for everything that can support it.** This covers wheelchair users, people with balance or mobility difficulties, and the large number of people who simply want to sit down.
- **One-handed modes.** Requiring two functioning hands excludes people, and it is rarely genuinely necessary — most two-handed interactions have a reasonable one-handed equivalent.
- **Reachability.** Do not require reaching above the head, to the floor, or behind the body. Offer an adjustable interaction height so content can be brought within a user's range.
- **Subtitles that work in three dimensions** — positioned at a readable depth and angle, legible against arbitrary backgrounds, and **with an indicator of which direction the speaker is in**, since a deaf or hard-of-hearing user loses the audio cue that would otherwise direct their attention (Module 11).
- **Colour and contrast**, as in any medium — with the added consideration that VR displays have their own contrast limitations and that god rays (Module 04) make high-contrast content worse. **VR accessibility is substantially less developed than flat-screen accessibility**, which is both a problem and an opportunity.

2 Evaluation

YOU ARE THE WORST AVAILABLE TEST SUBJECT

You know where everything is. You know which objects are interactive, what the control mapping is, and where the experience intends you to look — because you decided all of it.

You built the mental model you would be testing for. There is no way to un-know it, and no amount of care substitutes.

And you are maximally acclimatised to VR discomfort (Module 07), having spent more hours in a headset than any of your users will.

So your judgement is the least reliable available on exactly the two dimensions that matter most — usability and comfort. **This is not a failure of care or attention.** It is structural, it applies equally to everyone who builds VR, and the only remedy is to test on other people.

Claim you want to make	Appropriate instrument
'It is comfortable'	The Simulator Sickness Questionnaire or the shorter VRSQ, administered before and after (Module 07). Free, validated, and comparable to published work.
'It is immersive' / 'users feel present'	A validated presence questionnaire — Witmer–Singer, or Slater–Usch–Steed. Presence is subjective and these make it comparable.
'It is usable'	Task completion rate, time on task, error counts, and a think-aloud protocol for the reasoning behind failures.
'It is discoverable'	Time to first successful interaction, with no instruction. The single most informative usability number in VR.
'It performs well'	The frame time distribution, including the 99th percentile — never the mean (Module 01).
'It is responsive'	Physically measured motion-to-photon latency (Module 06), not an estimate from frame time.

WATCH THEM IN SILENCE

Give a person who has never seen your application the headset, say nothing beyond 'have a look around', and watch for sixty seconds.

Record every hesitation. Where did they look for something and not find it? What did they try that did not work? What did they not notice at all? What did they reach for first?

Resist explaining. The instant you say 'press the trigger', the session has stopped measuring discoverability and started measuring whether they can follow instructions — and you have learned nothing you did not already know. This is genuinely difficult to do and it is where the value is.

Six participants will find most usability problems, and the first two will find the majority of those. **This is cheap, it takes an afternoon, and almost nobody does it** — which is why so many VR applications have a first minute that nobody can get through.

3 Ethics

- **Physical safety.** Users cannot see the room they are standing in. Guardian and boundary systems exist for this reason and should never be worked around; designs that encourage large, fast, or backward movements are designs that will injure somebody.
- **Intensity.** VR experiences are substantially more affecting than screen-based ones — heights produce genuine fear responses, and threatening content produces genuine startle. **Including in ways users do not anticipate**, because their expectations are calibrated on screens. Warn in advance, and make exiting easy and obvious.
- **Data.** Eye tracking, hand tracking, and room scanning produce unusually intimate data. **Gaze in particular reveals attention, interest, and intent** at a resolution no other consumer sensor approaches — what someone looked at, for how long, and what they looked away from. This is the consideration most likely to matter over the next decade, and it is underdiscussed.
- **Harassment in social VR has a physicality that text-based harassment does not.** Proximity, gesture, and voice combine into something users report as qualitatively worse. Personal space tools and blocking exist in social VR because they were found to be necessary, not as a precaution.
- **Children.** Developing visual systems, unknown long-term effects, headsets built around adult IPDs (Module 03) that do not adjust far enough, and age ratings that are routinely ignored. **The honest position is that the long-term effects are not known**, which is itself a reason for caution rather than a reason to proceed.

WHERE THIS COURSE LEAVES YOU

You can account for every millisecond between a head movement and a photon, and explain why that accounting is physiological rather than aesthetic.

You can distinguish a rendering problem from a tracking problem from a perceptual problem — which, as in every other course in this program, is most of the practical skill. A juddering world, a swimming world, and a nauseating world have three different causes and three different fixes.

And you know that your own judgement of comfort and usability is the least reliable available. If this course teaches one thing that survives, it should be that — because it is the belief that most reliably prevents shipping something that makes people ill.

CSCE 641 and 647 render it; CSCE 649 moves it; CSCE 735 makes it fast enough. This course is where a person is placed inside the result, and the constraint stops being a frame budget and becomes a physiological one.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 12 (free)	Evaluation methodology, the standard instruments, and experimental design for perceptual claims.
Meta — VR design and accessibility guidelines (free)	The convention set of §1 and a growing accessibility section.
XR Access — accessibility resources (free)	The most active community work on §1's accessibility material, including guidelines and user research.
Nielsen — Why You Only Need to Test with 5 Users (free)	The basis for §2's claim about six participants. Short, and the argument generalises to VR.
Slater & Sanchez-Vives — Enhancing Our Lives with Immersive Virtual Reality (free)	A survey of what the medium can do, including the ethical questions in §3, from researchers who have studied them.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Audit three VR applications against the convention table in §1 and note every violation and its consequence.
- 2 Implement the full standard comfort settings set and justify each option from Modules 07 and 08.
- 3 Add height calibration and test your application seated, standing, and at a child's height.
- 4 Add a one-handed mode to an interaction that assumed two hands.
- 5 Implement 3D subtitles with a speaker direction indicator and test them with the audio muted.
- 6 **Run the silent-observation test** on six people, at least two of whom have never used VR. Record every hesitation and resist explaining.
- 7 Administer an SSQ before and after, and a presence questionnaire after, for the same six people. Report every score individually.
- 8 Measure time to first successful interaction for each participant, and identify the single change that would most reduce it.
- 9 Write the ethics section of a design document for your Project 2 application, addressing each item in §3 that applies.
- 10 **Project 2 is now due.** Submit the application, the frame timing data, the participant results in full, the optimisation measurements, and the account of something that worked for you and not for others.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name six VR interface conventions and say why each settled.
- 2 Why are comfort settings an accessibility feature rather than a courtesy?
- 3 Give five VR-specific accessibility requirements.
- 4 Why are you the worst available test subject, on which two dimensions?
- 5 Name the appropriate instrument for each of six claims you might want to make.
- 6 Describe the silent observation method and say what destroys the data.
- 7 How many participants find most usability problems?
- 8 Name five ethical considerations the medium raises, and say which is most likely to matter over the next decade.
- 9 What are the three categories of problem you should be able to distinguish?